Bioinformatics (COMP0082): -Omics and Systems Biology

Introduction

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What are these things and why do we need them?

Overview

- **Lecture 1:** Intro, background and overview
- **Lecture 2:** Experimental techniques in -omics
- **Lecture 3:** Basic algorithms in -omics analysis
- **Lecture 4:** Graph theory and applications to biological networks
- **Lecture 5/6:** Applications of ML in systems biology and -omics

What are -omics?

- Branch of molecular biosciences concerned with studying the entire complement of some entity or process

History

1850

Pasteur fermenting of glucose

Mendel and the peas ~1856

Discovery of Chromosomes 1880

Active enzyme extracts 1897

1900

“Genes” as points on chromosomes ~1910s

Complete Glycolysis Pathway 1930s

DNA *is* the genetic material ~1940s-50s

1950

Structure and encoding potential of DNA 1953

Full sequence of a Gene 1972

Sanger sequencing 1977

2000

OMICS 90s ->

2020

Gene theory

Metabolic networks as
a model of cell function

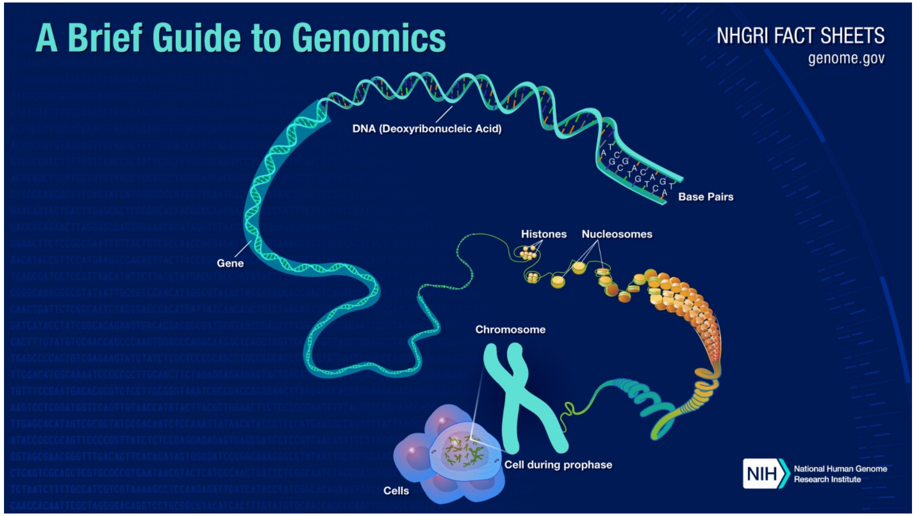
What are genes?

Chromosomes have both
nucleotide and protein

https://www.ncbi.nlm.nih.gov/pmc/articles/

Genomics

- Genome Etymology ~1926
- Genomics – 1986



Meaning: The study of the structure, function, and evolution of genomes.

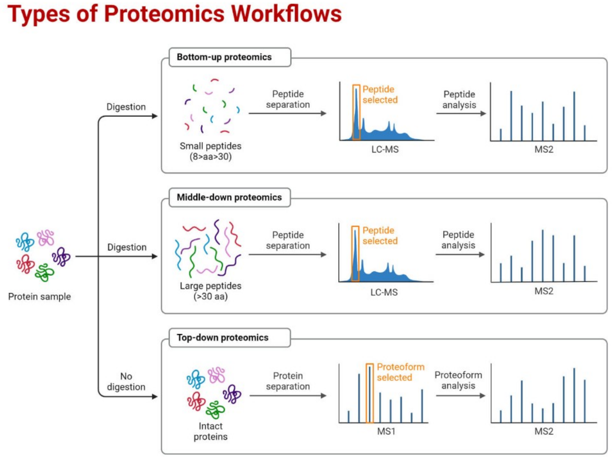
https://www.genome.gov/about-genomics/fact-sheets/A-Brief-Guide-to-Genomics

Genomics

Determining the complete DNA sequence of an organism

Year	Lab	Organism	Size
1977	Fred Sanger	phage virus Φ-X174	5000bp
1995	Celera	H.influenza	1.8Mbp
2003	Human Genome Consortium	Homo Sapiens	3Gbp

Proteomics



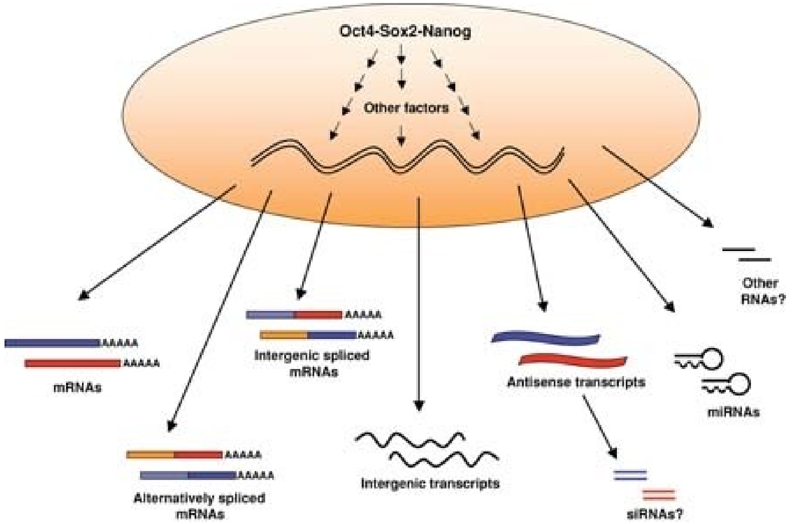
- Proteome - 1994
- Proteomics – 1995

Proteome meaning: The complete set of proteins an organism can produce

Proteomics: Large scale study and identification of proteins in biological/cell samples

https://microbenotes.com/

Transcriptomics

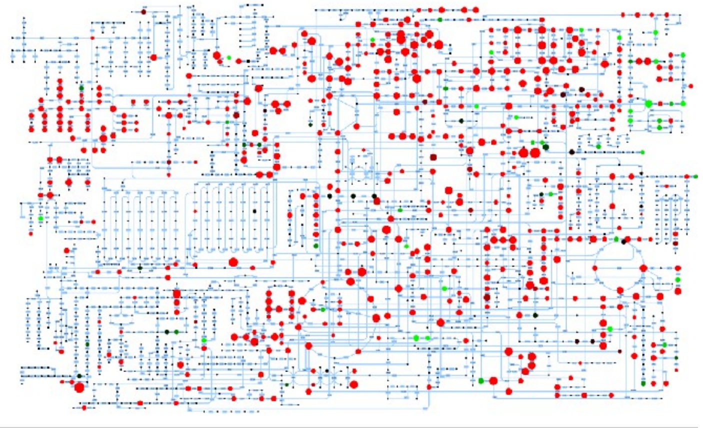


- Transcriptome - 1997
- Transcriptomics – (1979) 1995

Transcriptome meaning: The complete set of RNA transcripts and organism produces
Transcriptomics: The identification of all species of RNA transcripts in cell and tissue samples

https://bio.davidson.edu/genomics/2014/Kyaw/transcriptome.html

Metabolomics



- Metabolome - 1998
- Metabolomics – ‘40s – ‘90s -2000s?

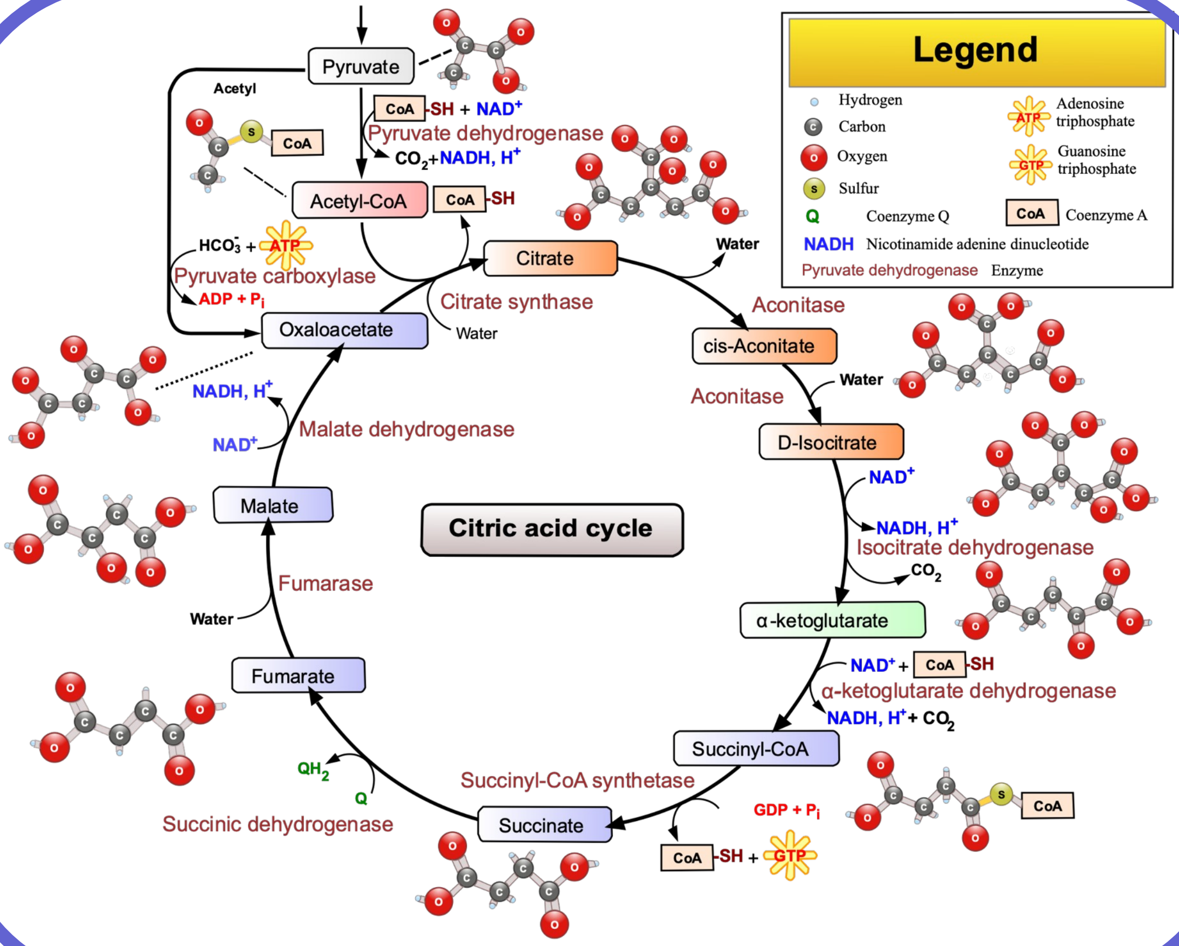
Metabolome meaning: The complete set of small molecule metabolites present in an organism

Metabolomics: study of all process that involve metabolites

https://imsb.ethz.ch/research/sauer/research/

Metabolomics

- Cataloguing and measuring the presence and concentration of all metabolites in a tissue



Wikimedia CAC

Cartesian view of biochemistry

- Animals are but machines

  - Reductionism: can we reduce living things to a list of parts?
  - Holism: Can we analyse things in their totality

Why are -omics?

- A means of enumerating all the parts
  - Transcriptome – all the transcripts
  - Genome – all the genes
  - Proteome – all the proteins
  - …

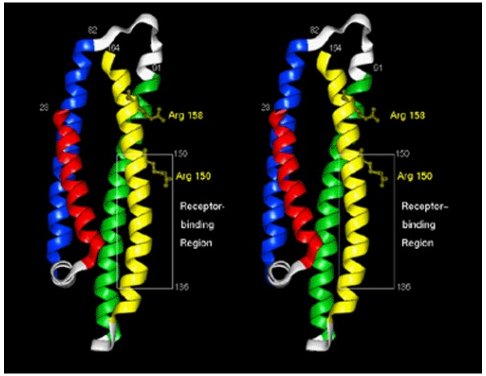
Why stop there?

Why stop there?

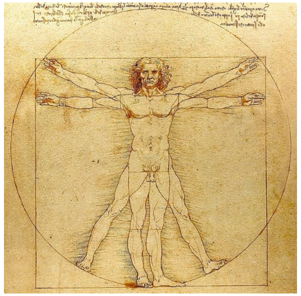
We want to take all the parts and reconstruct the systems

Systems biology

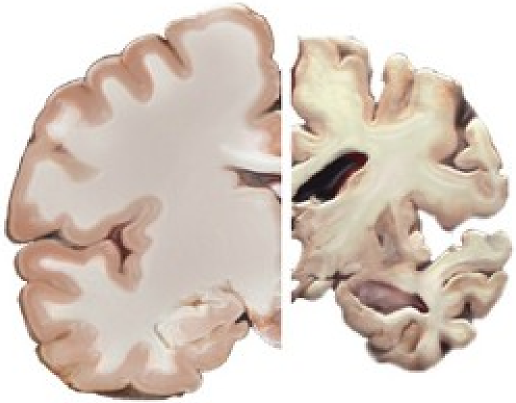
We want to understand complex biochemical systems



ApoE Rupp lab 1996

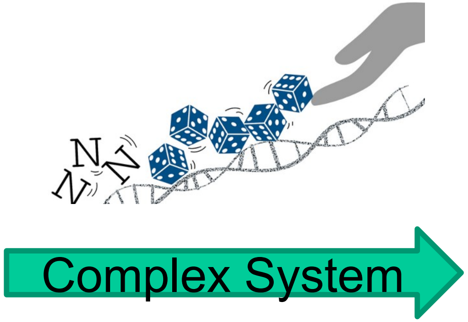


Complex System



We want to understand complex biochemical systems

Healthy



Melanoma

We want to understand complex biochemical systems

E-CELL

Whole cell Simulation (1999-2021)

https://www.e-cell.org/

https://pubmed.ncbi.nlm.nih.gov/10068694/

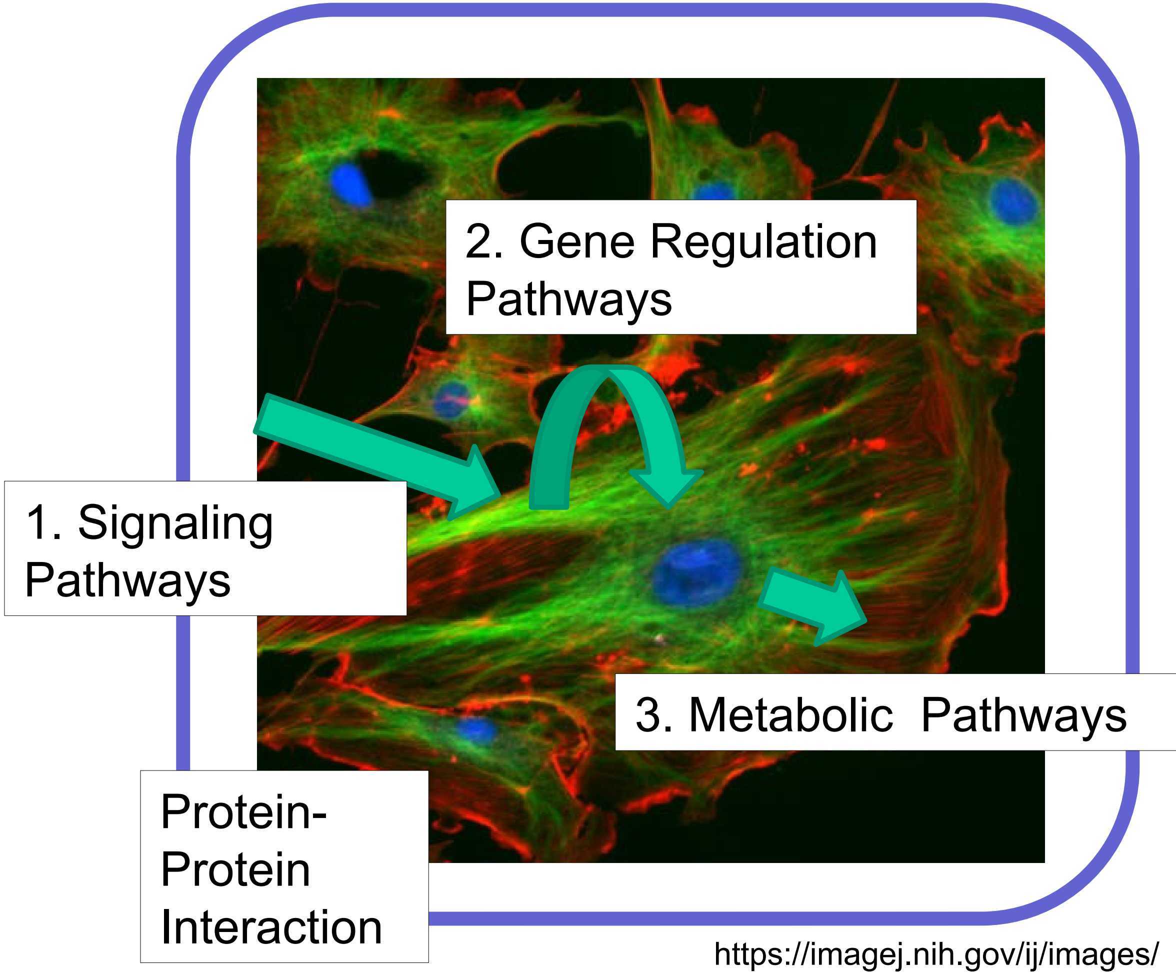
Systems Biology: a science of networks

Consider Insulin Signaling:

1. Insulin Stimulates a Liver Cell

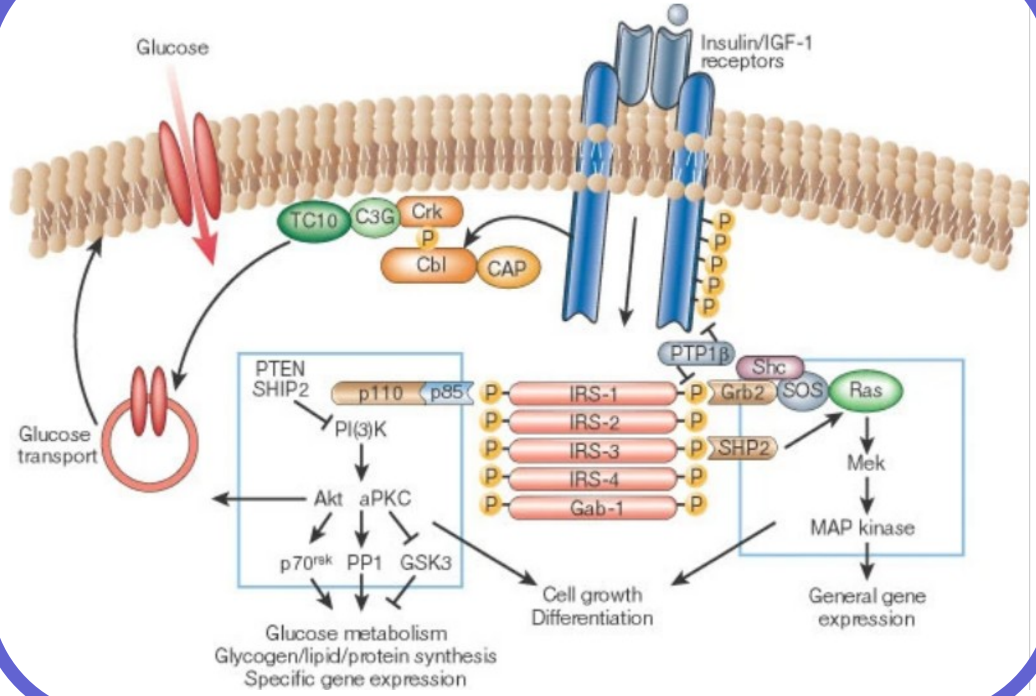
2. Turns on Genes for glucose uptake and processing

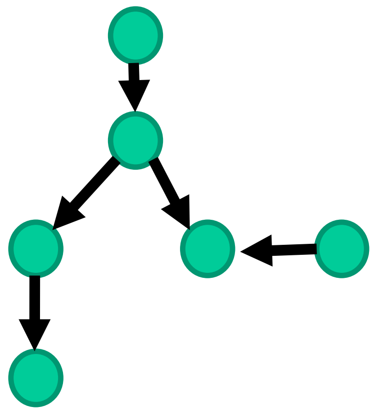
3. Glucose transformed to glycogen, fall in blood glucose levels



https://imagej.nih.gov/ij/images/

1. Signalling Networks





Nodes: Proteins
Edges: Interactions/Activations/Repression

Activation

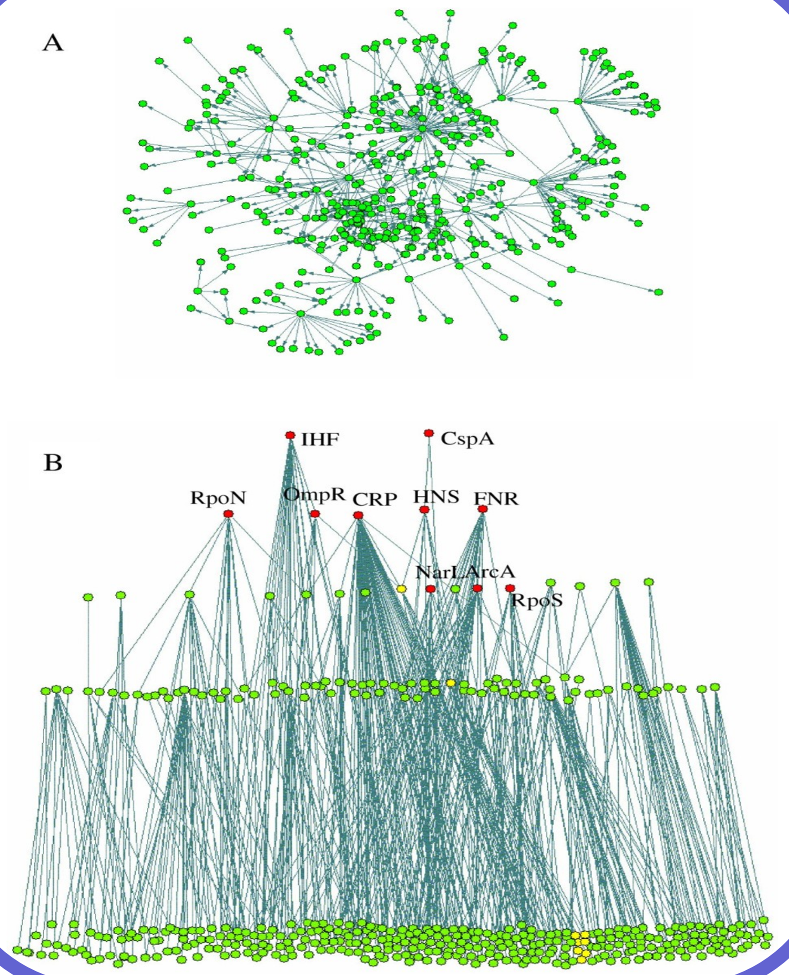
Repression

Insulin signalling and the regulation of glucose and lipid metabolism. Nature. 2001 Dec 13;4

2. Gene Regulation Networks

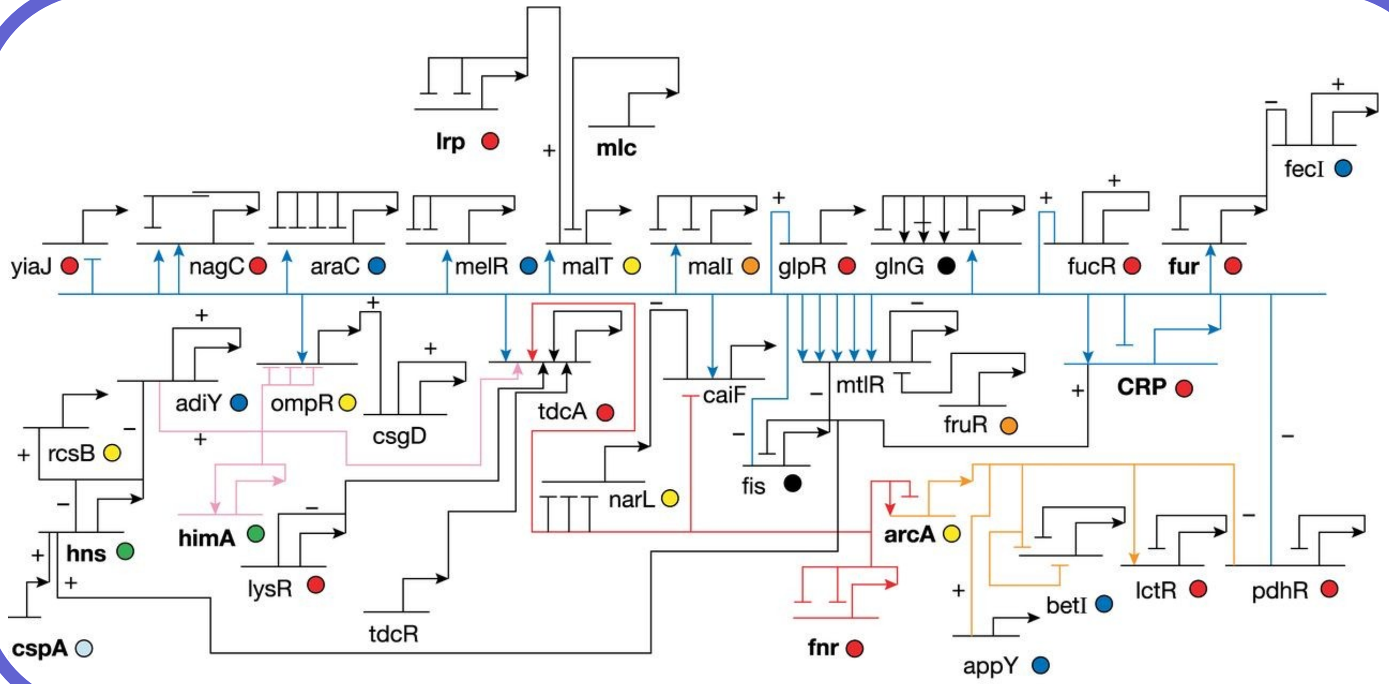
Nodes: Transcription Factors
Edges: Gene Regulatory Interaction

Insight: Gene regulation is hierarchical

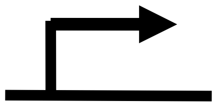


Hierarchical structure and modules in the Escherichia coli transcriptional regulatory network revealed by a new top-down approach (2004) BMC Bioinformatics 5: 1

Annotated Gene Regulation; E. coli



Activation Repression



Expressed Gene

- Homeodomain-like
- 'Winged helix' DNA-binding domain
- FIS-like
- Nucleic acid-binding proteins
- IHF-like DNA-binding domain
- C-terminal effector domain of the bipartite response regulator
- Lambda repressor-like DNA-binding domain

Charting gene regulatory networks: strategies, challenges and perspectives (2004) Biochem. Journal 381:1-12.

3. Metabolic Networks (Insulin)

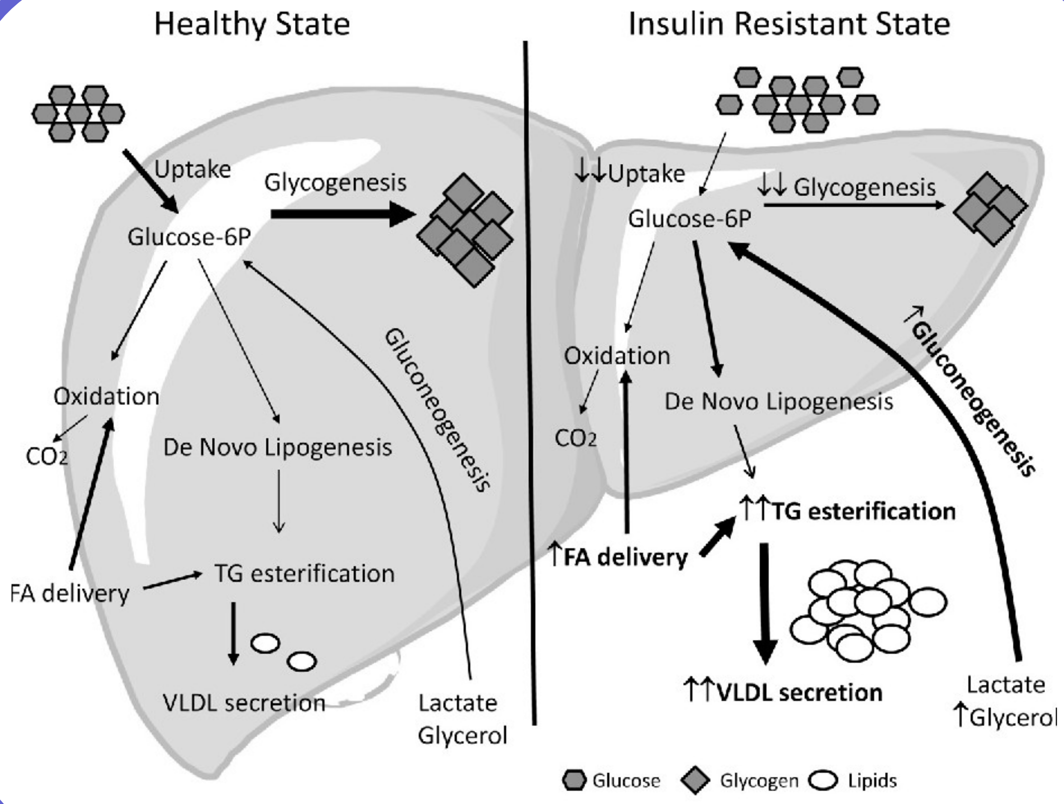


FIGURE 1. **Hepatic substrate flux in the fed state in healthy and insulin-resistant subjects.**

1. Signalling network with proteins (de-) phosphorylating other proteins to transmit a signal.

2. Gene regulation network with transcription factors binding to particular sequences in the genomic DNA to turn on/off other genes.

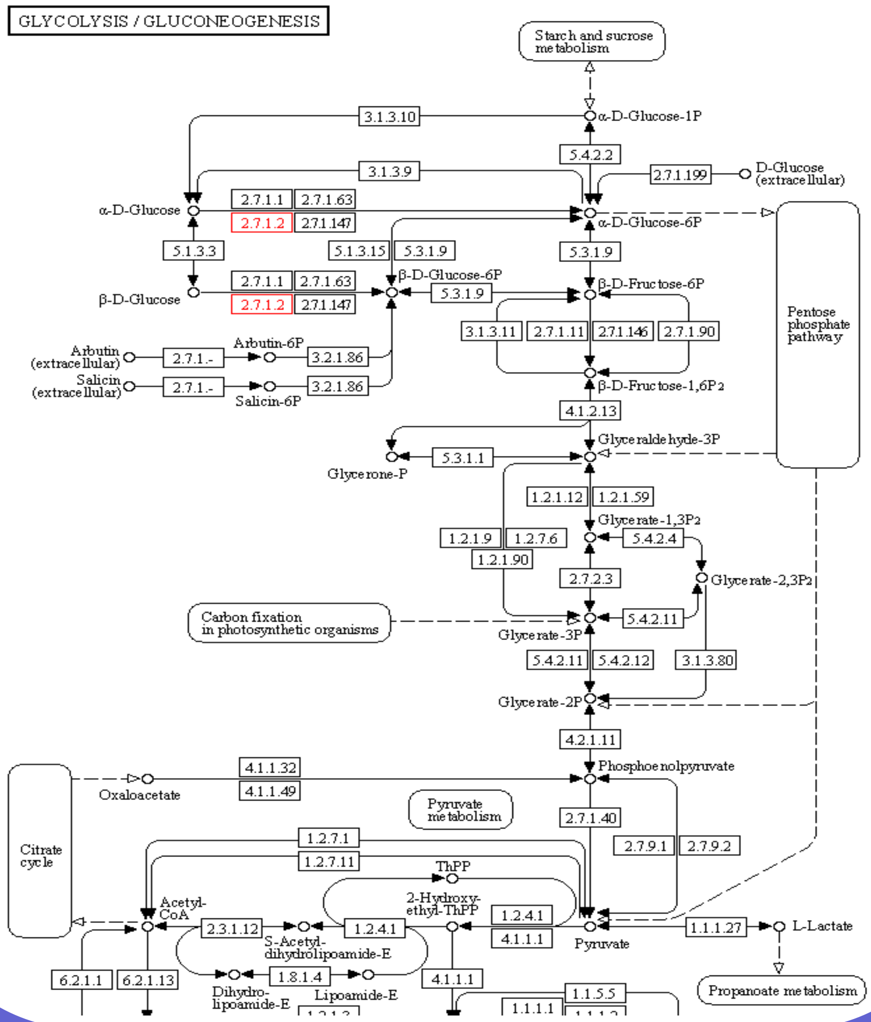
3. **This then increases/decreases the expression of particular enzymes which changes the processing of small molecules in metabolic pathways.**

"Pathway-selective Insulin Resistance and Metabolic Disease: The Importance of Nutrient Flux." Journal of Biol Chemistry 289 (2014): 20462 - 20469.

Metabolic Pathways (KEGG database)

Node: metabolite (small molecule such as glucose)
Edge: conversion of one metabolite to another via enzyme

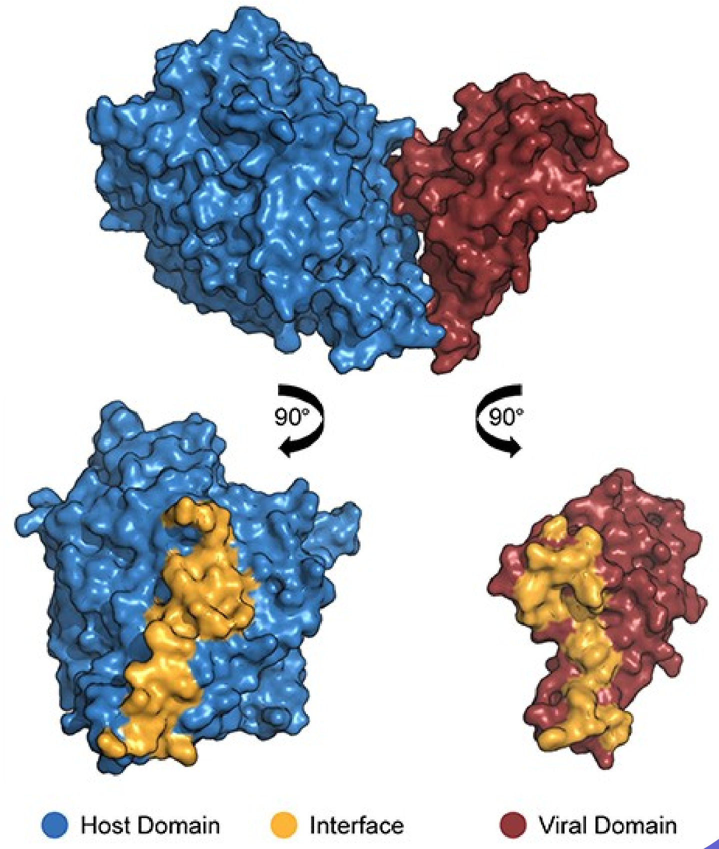
This shows a small part of glucose breakdown in a bacteria The concentration of enzymes are regulated via the gene regulatory network (i.e. all the pathways are linked)



4. Protein-Protein (P2P) Interaction Network

Proteins interact with each other

- (De-)phosphorylation or other modifications of amino acid residues (in signaling pathways)
- Allosteric activation (transcription factors, enzymes)
- Inhibitory binding (covering the active site)
- Physical closeness of enzymes along metabolic pathway. – Etc.

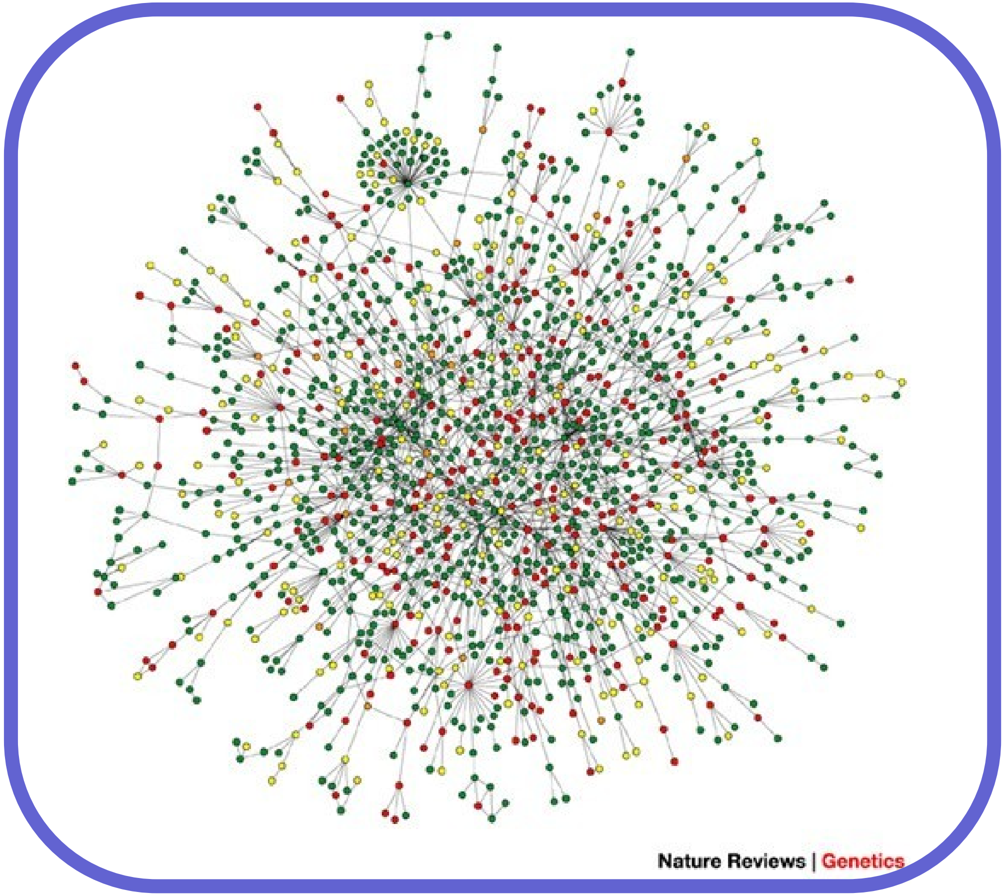


Protein–Protein Interactions in Virus–Host Systems Front. Microbiol., 17 August 2017 Sec. Virology

Protein-Protein (P2P) Interaction Networks for Yeast

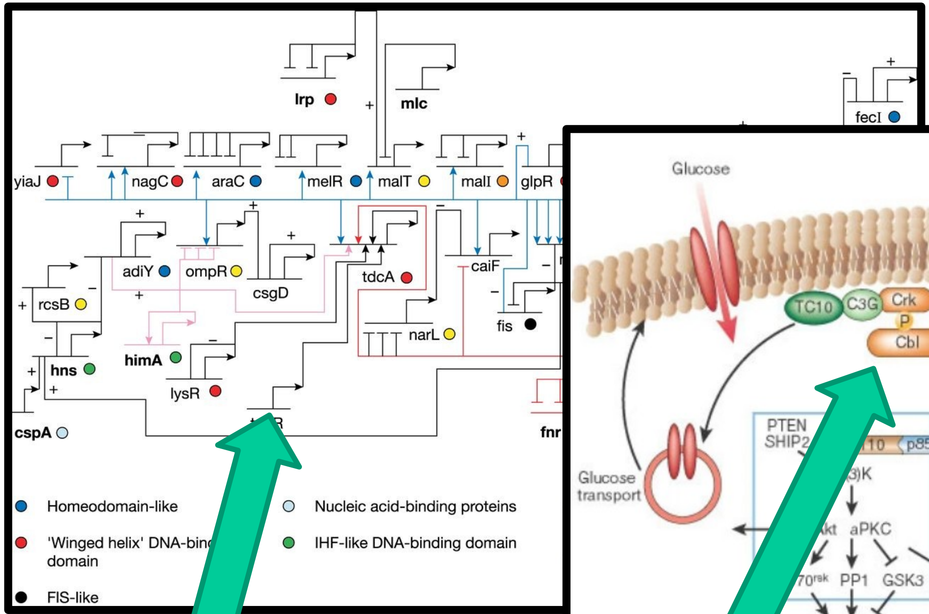
Nodes: Protein
Edges: Physical interaction

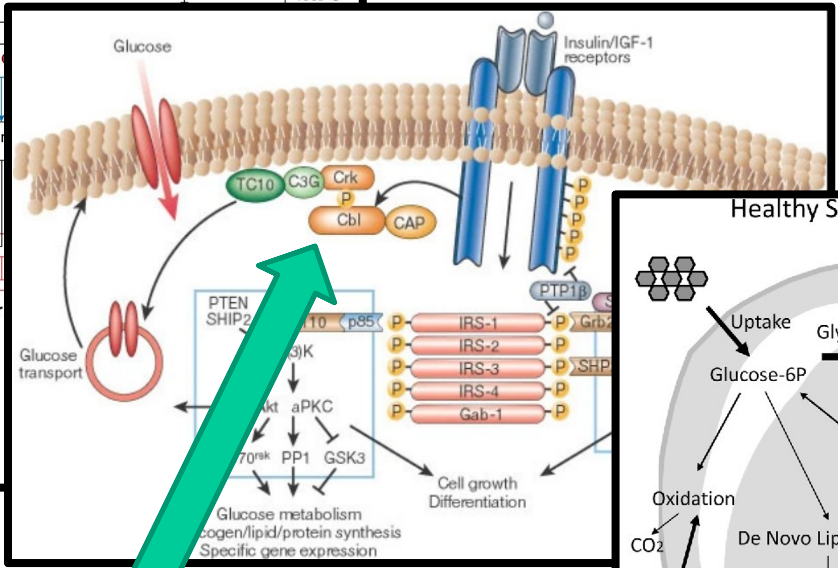
- robust "scale-free" network
- Hub nodes are general red where deletion is lethal
- Provides an indication of 'functional modules'.



Network biology: understanding the cell's functional organization (2004) Nature Reviews Genetics 5:101-113.

How are network data and *–omics* data related?





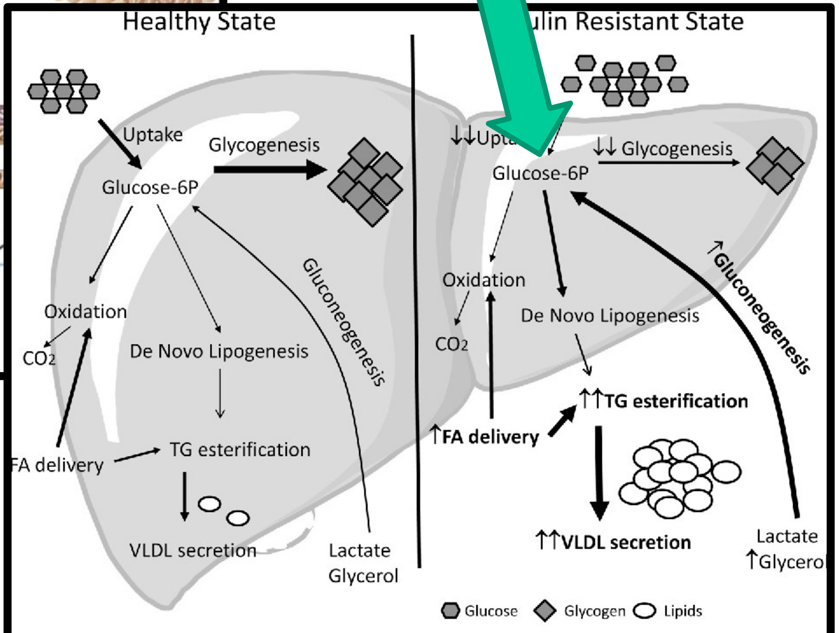


FIGURE 1. **Hepatic substrate flux in the fed state in healthy and insulin-resistant subjects.**

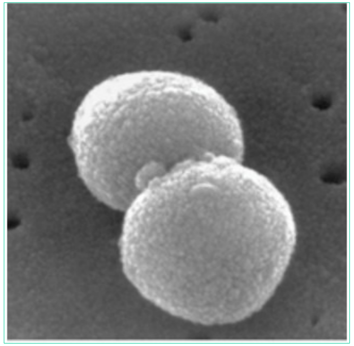
Metabolomics

Transcriptomics -> Genomics

Proteomics

Transcriptomics Example

Motivating Example
Streptococcus pneumoniae

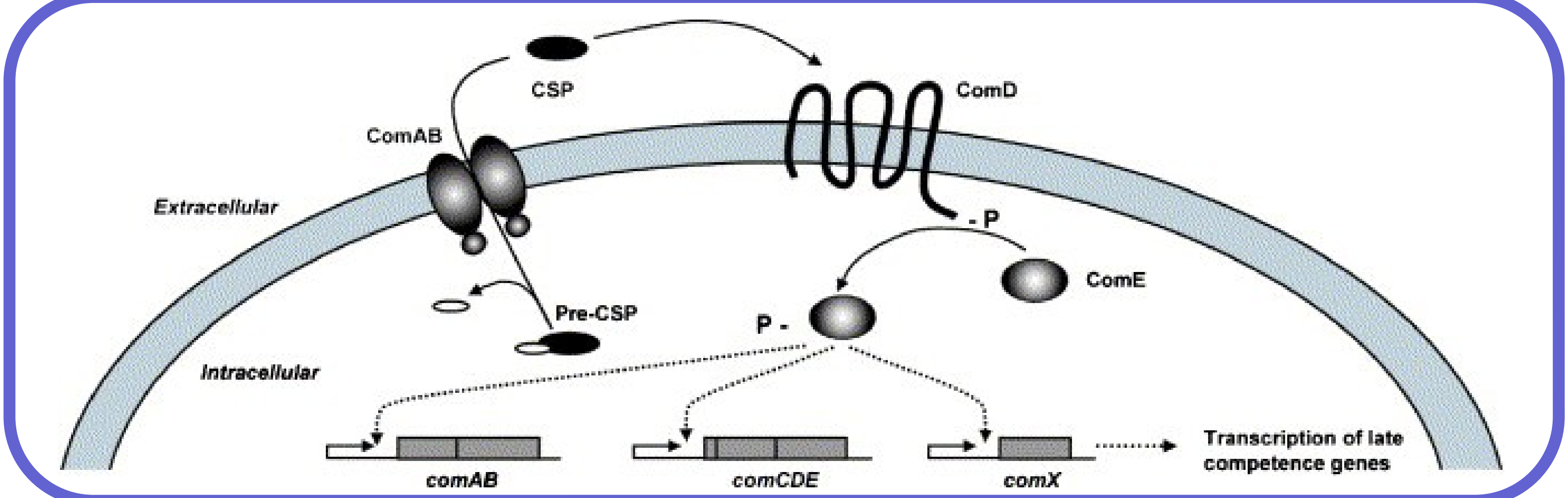

Maidenlab.zoo.ox.ac.uk

- Isolated by Louis Pasteur and US Army physician George Sternberg in 1881
- "Pneumococcus" due to its role in pneumonia

- Part of the normal upper respiratory tract flora so you all have this bacteria right now !

- Only occasionally does it become pathogenic and cause disease … with 1.6 million deaths annually … mostly under 5s and elderly …

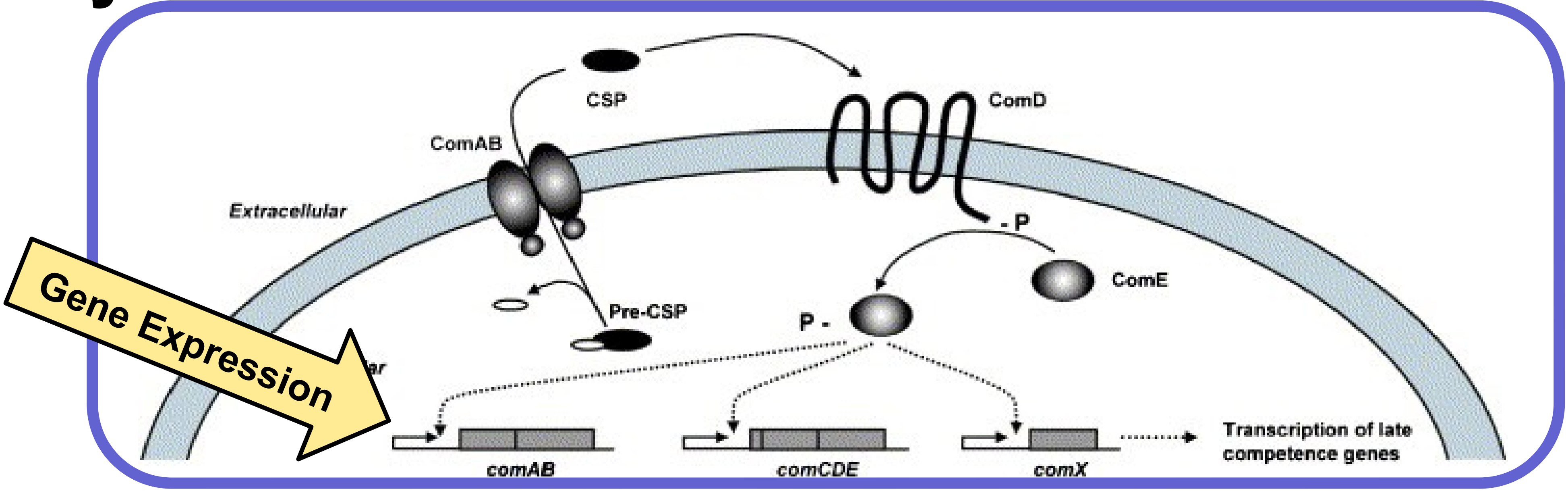
How does it become pathogenic?



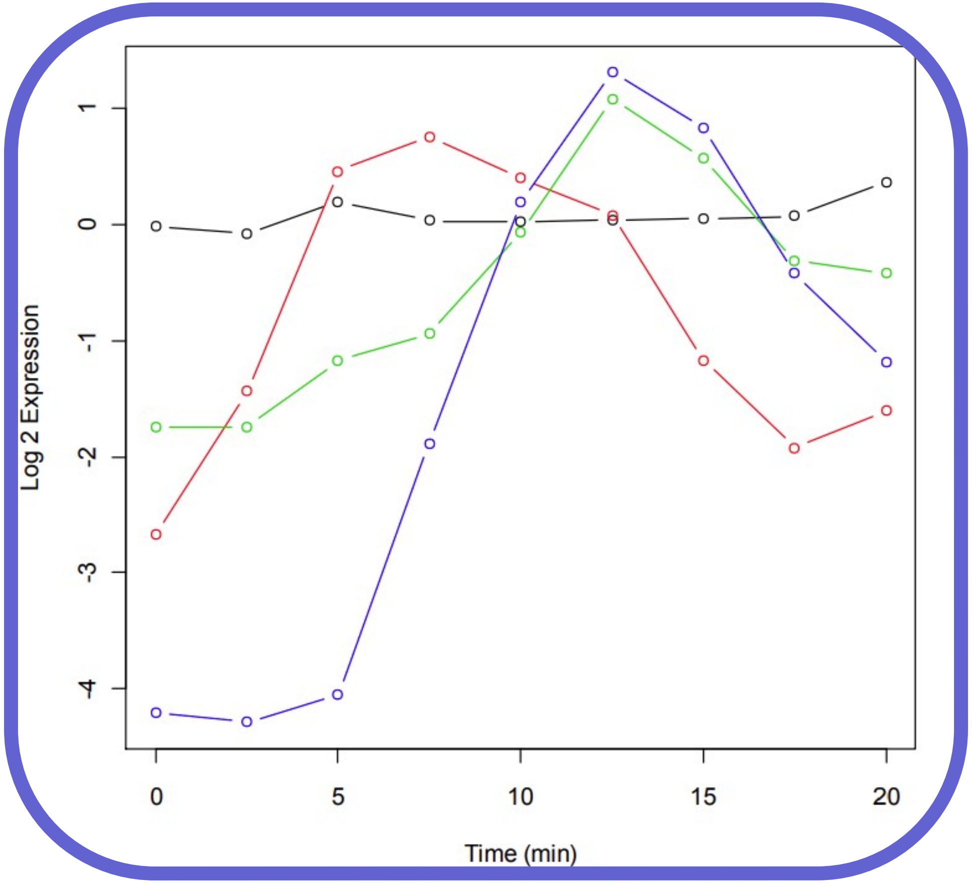
- Competence stimulating peptide (CSP) to detect density of bacteria cells
- This is used (by the bacteria) to time when it becomes pathogenic
- We can make use of this to design peptides to stop this process

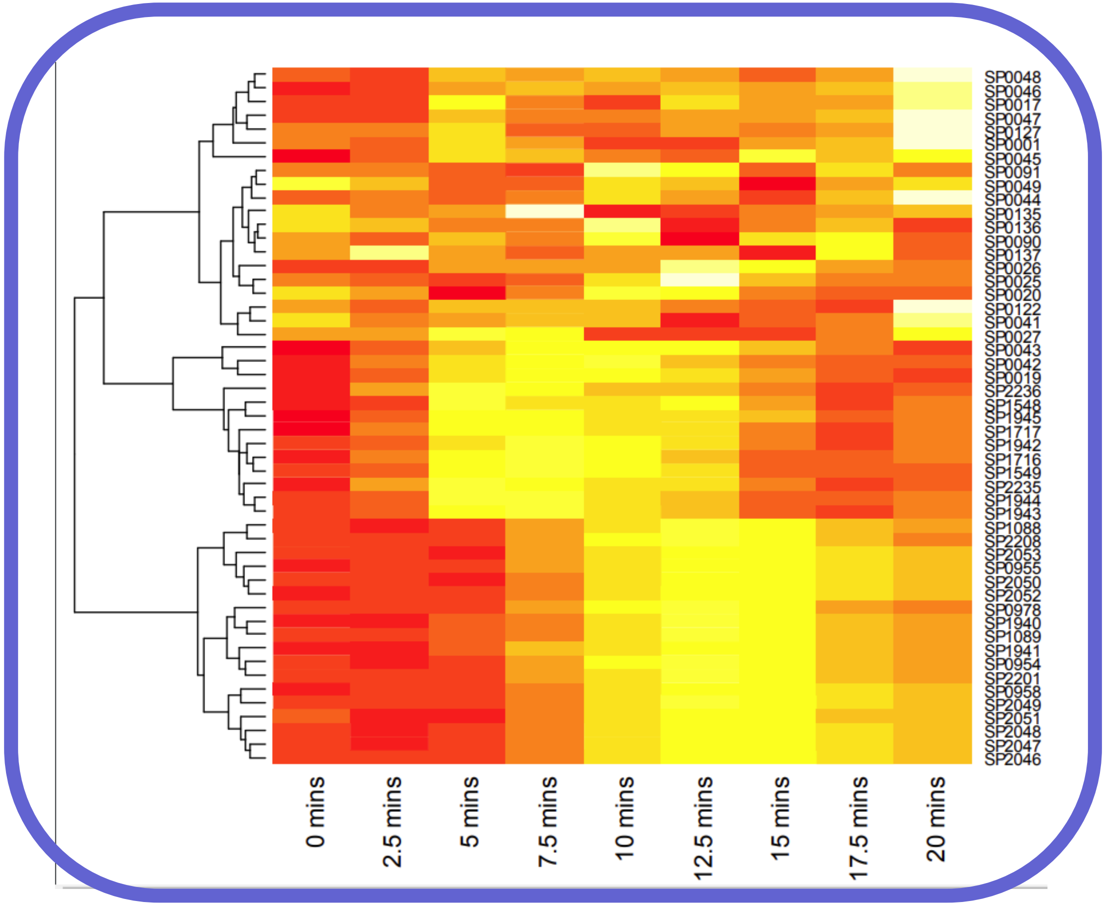
Modeling the regulation of the competence-evoking quorum sensing network in *Streptococcus pneumoniae* Biosystems, 2007, 90(1):211-23

We must understand complete biological system



Modeling the regulation of the competence-evoking quorum sensing network in *Streptococcus pneumoniae* Biosystems, 2007, 90(1):211-23





Identification of competence pheromone responsive genes in Streptococcus pneumoniae by use of DNA microarrays (2004) Peterson et al. Mol Microbiol. 51(4):1051-70.

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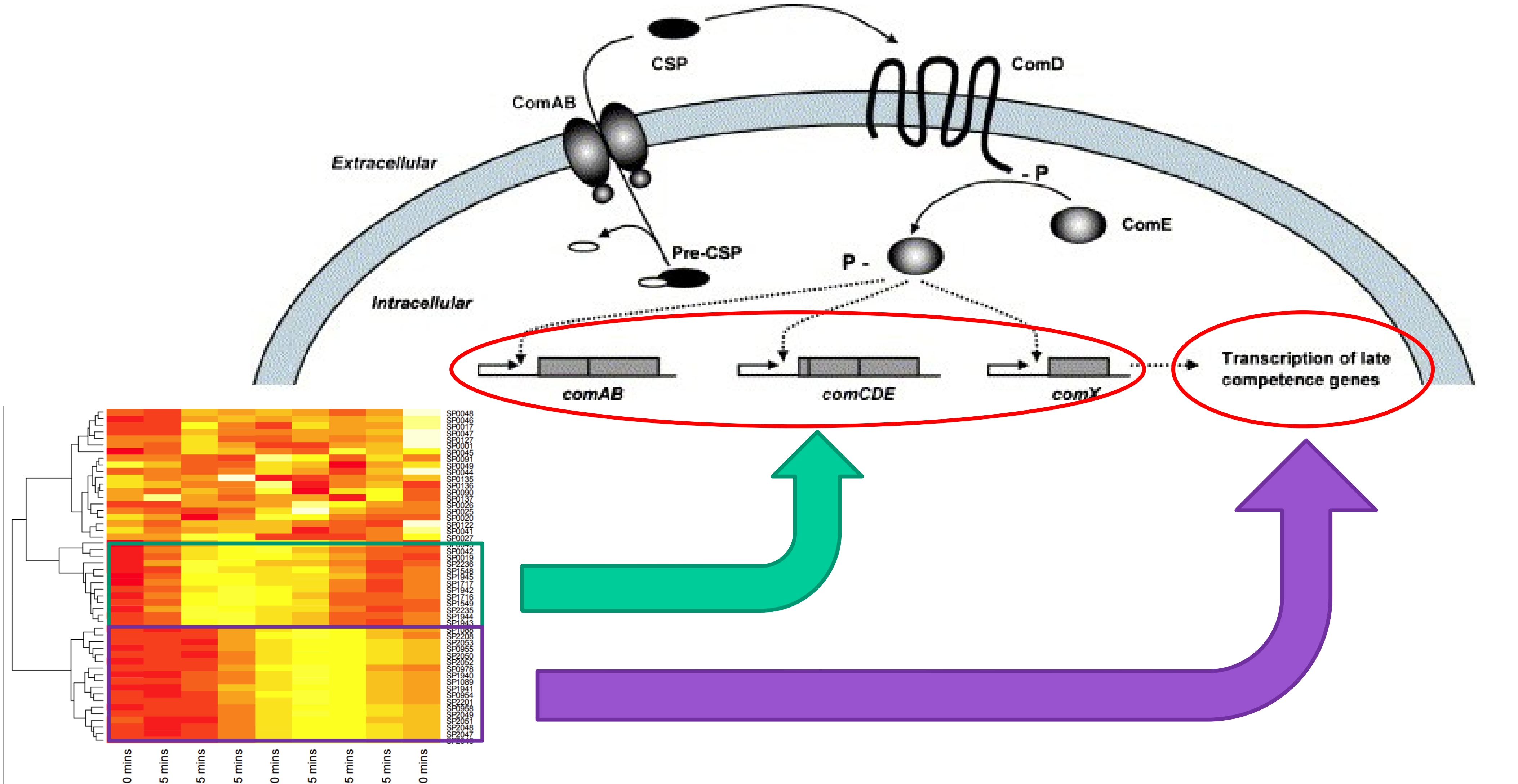
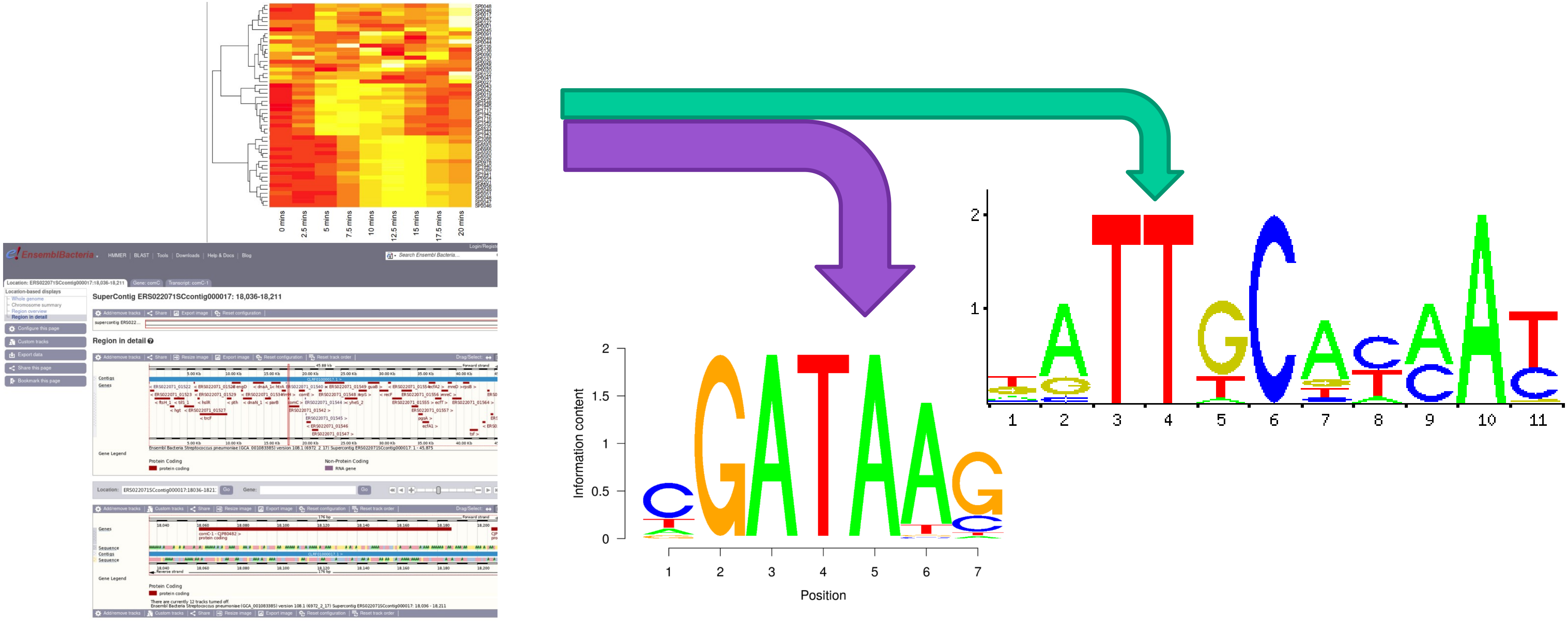
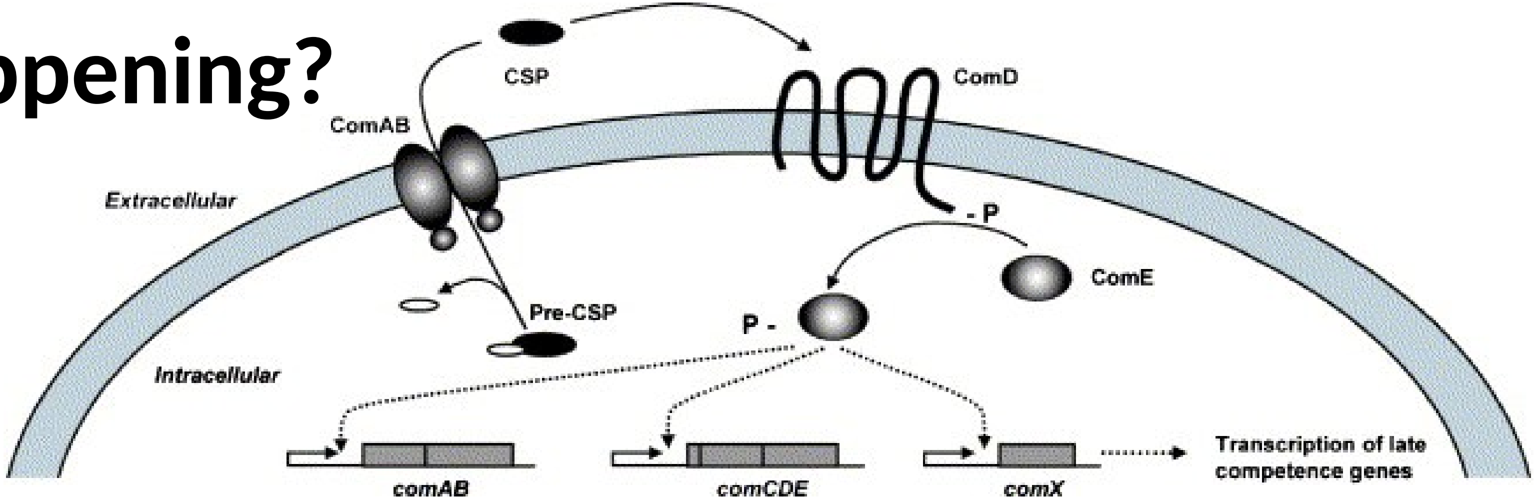
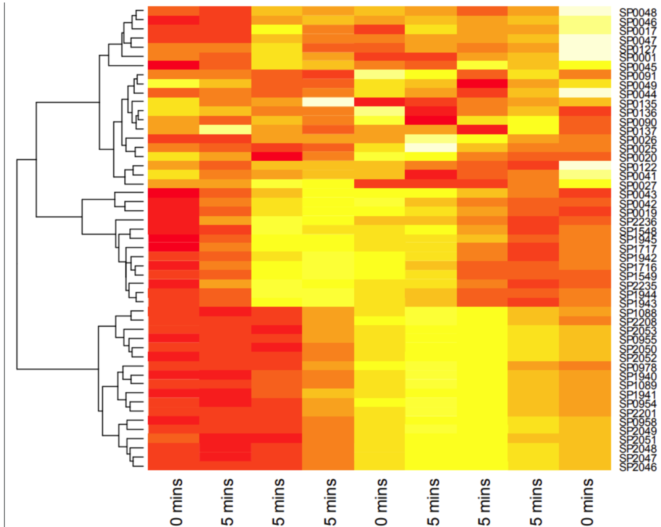

CSP
ComD
ComAB
Extracellular
- P
ComE
Pre-CSP
P -
Intracellular
comAB
comCDE
comX
Transcription of late competence genes

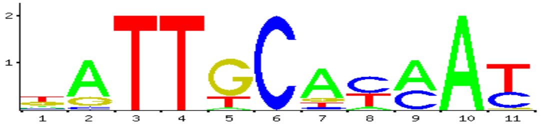
Align DNA sequences upstream of the genes. Apply “motif detection” shared DNA patterns



What is happening?
Questions?







Summary

- To understand most complex biological systems (i.e diseases)
  - Need an understanding of the components (genes, proteins, metabolites, etc.)
  - The interactions of these components in terms of networks of pathways – "systems biology".

- We introduced a number of high-throughput –omics technologies
  - genomics for getting a genes 'parts list'
  - transcriptomics/proteomics/metabolomics for component flux

- We also introduced biological network data
  - signaling pathways
  - gene regulation networks
  - protein-protein interaction networks
  - metabolite networks